Token Usage, Compaction & Fertility

Deep Dive — Languages · Formats · Custom ADE Languages · Hard Limits

What actually shrinks the channel · What models can still use · What ADE/harnesses can enforce

Edition: 2026-07-23 · **Schema:** `ade.compaction-research/v1`
Method: Parallel research (compaction science · fertility · custom DSL limits) + primary-source spot checks
Companions: *ADE Tokenomics* (I/O playbook) · Continuity `ade.handoff/v1` · Effort B0–B4
Stance: Fertility ≠ quality · compaction ≠ free lunch · inventing “machineese” is usually the wrong bet

0. How to read this paper

Label	Meaning
HIGH	Vendor docs, peer-reviewed / arXiv with clear method, or multi-source corroborated
MEDIUM	Strong first-party engineering blog or single-study ranking
ESTIMATE	Workload-dependent; measure on ADE gold before treating as law

Two different compressions people confuse:

Kind	Question	Owner
Fertility	Same *text* → fewer/more **tokens** (tokenizer)	Frozen for API models
Compaction	Same *mission* → fewer tokens **kept in the window** (drop/summarize/externalize)	Harness / ADE

Custom "ADE languages" mostly try to hack fertility. Harnesses win by hacking **compaction + structure**.

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1. Executive verdict

1. **Best compaction for coding agents is hybrid (HIGH/ESTIMATE):** clear/mask stale **tool blobs** early; keep recent turns + paths/errors **verbatim**; at semantic boundaries (or ~50–80% window) write a **structured capsule**; put durable state on **disk** (`.ade/`), not in chat immortality.
2. **Observation masking often matches or beats LLM summarization** on SWE-bench-style agents (**HIGH** — Complexity Trap, arXiv:2508.21433). Don't pay for a summary when a stub + re-fetch works.
3. **Structured sections beat freeform prose** for continuity probes (**HIGH** — Factory / Cursor / Anthropic pattern). Ratio alone is a bad quality proxy (~98% token removal can still lose artifacts).
4. **Natural-language fertility (English-centric BPE):** English densest; Chinese/Japanese ~1.9–2.3×; some languages **12–15×** (**HIGH** — Petrov et al., NeurIPS 2023). That is **channel cost**, not "write ADE docs in Chinese to save tokens."
5. **Formats as LLM text:** TSV/CSV ≫ compact JSON ≫ YAML/TOML ≫ pretty JSON ≫ XML/HTML soup (**MEDIUM**). Pay for XML/JSON **delimiters** when reliability > fertility.
6. **Code fertility (directional, MEDIUM):** Python densest among common stacks; Rust/C/Go denser in *characters* but often **worse** in tokens (OOD syntax + verbose types). Prefer **diffs / hunks**, not "rewrite the module in denser language."
7. **Inventing a private ADE language for token savings is a weak bet (HIGH).** Frozen tokenizers fragment rare glyphs; models expand/explain opaque codes; SQL/diff/JSON work because training is saturated. Treat "ADE language" as **contract syntax** (capsule schema, path index, catalog IDs) — not steganographic BPE.
8. **Hard limit:** You cannot invent new BPE merges for Claude/GPT. Soft neural codecs (GIST/ICAE-class) need **matched open-weight** train/serve pairs — not pasteable into closed APIs.
9. **ADE today:** Continuity capsules + soft system budgets + Effort output caps — **strong thrift spine**; missing auto **history** compaction, tool-blob clearing policy, and measured fertility gold.
10. **Next ADE bets (ranked):** tool-result clearing → structured capsule @ boundary/70% → write-before-compact to `.ade/` → SelfCompact-style rubric tool → fertility bench — **not** a private cipher.

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2. Token usage in agent loops (why compaction exists)

Each tool round typically **re-sends** system + history + prior observations:

```
occupancy_shape ≈ O(rounds × growing_context)
```

Stale tool blobs and dumps become a **permanent tax**. Compaction is how the harness resets that tax without abandoning the mission.

Cached tokens still occupy the window — caching changes price, not capacity (*ADE Tokenomics*).

Three-tier memory (production consensus, MEDIUM–HIGH):

```
Tier 0  Working context     — recent turns, active paths, live tool results
Tier 1  Session compact     — structured capsule / anchored summary
Tier 2  External memory     — .ade/goals, leases, NOTES, verify evidence, capsules on disk
```

Write durable facts to Tier 2 **before** Tier 1 compresses (write-before-compact). Otherwise compaction deletes what you never externalized.

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3. Compaction science — mechanisms & evidence

3.1 Mechanism taxonomy (**HIGH**)

Mechanism	What it does	Fidelity profile
Observation masking / tool clearing	Stub or drop old tool results; keep reasoning/actions	Near-lossless for *chain*; lossy for blob contents (re-fetchable)
Abstractive summary	Rewrite history → prose/sections	Lossy; drift across regenerations
Anchored iterative merge	Extend a fixed structured doc; don't regenerate from scratch	Less drift than full rewrite (**MEDIUM–HIGH**, Factory)
Opaque vendor compact	Encrypted carry-forward item	Not inspectable (OpenAI Responses)
Extractive token prune	Keep high-utility tokens (LongLLMLingua-class)	Strong for RAG/QA; transfer to agents **MEDIUM**
Fixed memory overwrite	Slot rewrite (MemAgent-class)	Near-lossless on trained QA; needs training
SelfCompact	Model invokes compaction **tool** + **rubric** for when	Adaptive timing; training-free scaffold

3.2 When to compact — threshold vs decision (**HIGH**)

Fixed token thresholds (~50–80% window, or vendor defaults like Anthropic ~150k / min 50k) are **structure-blind**: they can fire mid-derivation and force expensive reconstruction.

SelfCompact (arXiv:2606.23525): compaction **tool** + rubric (*fire* when sub-task resolved / converging; *suppress* mid-derivation or stuck). Reported: matches/exceeds fixed-interval quality at **30–70%** lower per-question token cost; up to **+18.1** points on math vs no-summary baseline across six benches / seven models.

Implication for ADE: prefer **semantic boundaries** (task claim complete, verify passed, Suggest→Apply handoff) *plus* a safety threshold — not threshold alone.

3.3 Measured quality (selected **HIGH**)

Finding	Source
Observation Masking ≈ or > LLM-Summary on SWE-agent / SWE-bench Verified (e.g. Qwen3-Coder-480B **54.8%** mask vs **53.8%** summary; Gemini Flash thinking mask **36.4%** vs summary **31.4%**)	Complexity Trap, arXiv:2508.21433
Cursor RL self-summary: **−50%** compaction error, summaries ~~**1/5** tokens (~~1k vs >5k prompt baseline) at 40k/80k triggers	Cursor self-summarization (2026)
CompactionRL: GLM-4.5-Air **+7.0** SWE-bench Verified under fixed budgets	arXiv:2607.05378
ACON guideline optimization: **26–54%** peak-token cut while preserving success	arXiv:2510.00615
Anthropic context editing **+29%**; editing+memory **+39%**; 100-turn search **−84%** tokens	Anthropic context management (2025)
Factory continuity probes: structured iterative > regenerate; artifact scores still weak (~2.2–2.5/5) for all vendors; ~**99%** ratio ≠ quality	Factory evaluating compression (2025)
LongLLMLingua: up to **+21.4%** NQ with ~**4×** fewer tokens (RAG/QA)	ACL 2024

Rule: Manage degradation (sometimes *improve* vs raw rot). Never claim compaction is quality-neutral without measurement.

3.4 What to keep vs drop (coding agents — HIGH pattern)

Keep (verbatim or structured)	Drop / clear / externalize
User goal / eng-goal id	Full file dumps already applied
Decisions & constraints	Redundant tool stdout
Failing tests / error excerpts	KB articles, draft essays
Paths / lease ids / owned_paths	Stale search hit lists
Next steps / open questions	Processed observation blobs (re-fetchable)
Recent N turns / last accessed files	Entire peer-agent transcripts

Claude Code-style pattern: summarize critical details + keep a **small** set of recently accessed files.

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4. Fertility — languages & formats

4.1 Fertility ≠ quality (HIGH)

EMNLP 2024 — Schmidt et al., *Tokenization Is More Than Compression* (arXiv:2402.18376): PathPiece minimizes corpus token count; **no clear CTC** ↔ **accuracy** link across 64 trained LMs. Pre-tokenization and vocab construction matter more than "fewest tokens."

Zouhar et al., ACL 2023: Rényi efficiency correlates with MT quality in some settings; **LREC-COLING 2024** shows Rényi can be **gamed** while quality drops.

ADE rule: Optimize fertility only inside a **quality floor** measured on gold.

4.2 Natural languages (English-centric BPE — HIGH)

Relative fertility vs English on FLORES-style setups (Petrov et al., NeurIPS 2023; GPT-4/`cl100k` line):

Language	≈ × English tokens
English	1.00
Portuguese / Spanish / German / French / Italian	~1.5–1.6
Chinese	~1.9
Japanese	~2.3
Vietnamese	~2.5
Arabic	~3.0
Extremes (some low-resource)	**12–15×**

Ahia et al., EMNLP 2023: up to ~**5×** among 22 languages; over-fragmented languages also get worse utility.

Character-level models flip the story (CANINE can make Chinese cheaper than English) — irrelevant for ADE's API frontier path.

Do not: translate product English → Chinese to "save tokens." You pay fertility *and* often quality/OOD. **Do:** keep ADE operator language = language the **model is strongest at** for coding (usually English for current frontier coding SKUs).

4.3 Code languages (directional — MEDIUM)

No Petrov-class peer paper; cross-lang token-density measurements suggest approximate order for *same task*:

Language	≈ relative tokens	Note
Python	1.00	Dense, high BPE support
JavaScript	~1.3	
TypeScript / Java	~1.45	
Go / Rust	~1.55–1.6	Types + syntax can fragment
C	~1.8	

Harness implication: Don't rewrite Rust→Python for tokens. Emit **unified diffs / JSON Patch**, not whole files. Fertility gains from "language swap" are dominated by **I/O shape**.

4.4 Data & markup formats (MEDIUM)

Format (as LLM text)	Density	Use when
TSV / CSV	Best for flat tables	Tabular tool results
Markdown tables	Very good	Human+model readable tables
Compact JSON	Good	Nested structured state
YAML / TOML	OK–worse than compact JSON (~15–25% on nested)	Config familiarity
Pretty-printed JSON	Worse	Debug only — don't rebill pretty
XML / HTML tags	Verbose	**Deliberate** boundaries (Anthropic-style)
Raw HTML soup	Worst	Strip/clean first

MessagePack / protobuf binary	Not for paste	Decode to compact text first

Reliability tax: XML/JSON tags cost tokens but reduce parse ambiguity. ADE capsules should prefer **compact JSON or tight tagged sections** — not invent sigils.

4.5 Cross-tokenizer myths

Claim	Verdict
Newer OpenAI tokenizers always use *more* tokens	**False** for many non-EN scripts (GPT-4o: large *reductions* vs prior)
Claude +~30% vs GPT-4o on some Python	**MEDIUM** third-party; measure yourself
"+30% Opus vs prior" blog claims	**LOW** until vendor-verified

Always count with the **same** provider tokenizer as inference.

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5. Could ADE invent its own language?

5.1 What people hope for

A dense "machineese": short glyphs, stego codes, private BPE — so every turn ships fewer tokens with equal meaning.

5.2 Hard limits (**HIGH**)

1. **Frozen tokenizer** on hosted APIs — you cannot add merges or reserved "ADE tokens."
2. **Rare strings often *increase* fertility** — unknown glyphs → byte fallback → more tokens, not fewer.
3. **Training support:** SQL, unified diff, JSON, regex, SVG paths work because corpora are saturated. Private ciphers are **OOD**.
4. **Expansion tax:** models "explain" opaque codes → **output** tokens explode (often the expensive side).
5. **Window still counts tokens** even if humans can't read them.
6. **Soft neural codecs** (GIST, ICAE, 500xCompressor-class) live in continuous/KV space with **matched training** — not pasteable into Claude/GPT chat.
7. Telegraphic English / BabelTele-class densification tops out around ~**2–3.5×** with fidelity claims that are **pair-dependent (MEDIUM)** — not a free 10× codec.

5.3 Feasibility matrix for ADE

Rank	Option	Call	Why
1	Catalog + ID refs (skills/rules by id)	**Best**	Already ADE DNA (T0–T3)
2	Structured handoff capsules (`ade.handoff/v1`)	**Best**	Continuity shipped; tighten sections
3	Diff-only / hunk I/O	**Excellent**	Pretraining saturated
4	Path / symbol indexes	**Excellent**	Replace dumps
5	Compact JSON capsules + verify pointers	**Excellent**	Fertility + parseability
6	Mild telegraphic rewrite inside known English	**Conditional**	~2×; test fidelity
7	Train open-weight compressor↔consumer	**Niche**	Only if ADE hosts both ends
8	Private stego / invented BPE language	**Avoid**	OOD + expansion + no merge control

5.4 What “ADE language” *should* mean

Contract syntax, not a cipher:

```
ade.handoff/v1 {
  goal_id, decisions[], failing[], paths[],
  next_steps[], verify_cmds[], lease_ids[]
}
```

Plus wire habits models already know: **unified diff**, **JSON Patch**, `skill:<id>` activation, `path:` indexes.

Optional later: open-weight **session compressor** that emits the same capsule schema — still not a new natural language.

5.5 Arguments FOR a custom language (steelman)

- Multi-agent handoffs could share a closed wire format with harness validation + NL escape hatch (**LOW** until measured).
- Soft codecs are real science — if ADE someday ships a local model pair (**MEDIUM** for that path only).
- Mild abbreviation conventions inside English (TE-class) can help if evaluated (**MEDIUM**).

5.6 Arguments AGAINST (decisive for API ADE)

- Structure already available in English+JSON+diff beats rare glyph soup (**HIGH**).
- Expansion and OOD destroy the savings (**HIGH**).
- Engineering time is better spent on **masking + capsules + boundaries** (see §7).

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6. Real limits — LLMs · ADE · harnesses

Layer	Can do	Cannot do
LLM (API)	Understand saturated formats; summarize; follow schemas	Learn your private cipher without fine-tune; escape context rot forever
Tokenizer	Fixed fertility map	Accept ADE-invented merges
Harness	Mask blobs; enforce schemas; externalize; Effort caps; Continuity	Make 200k window attend perfectly; bill fewer tokens than provider counts
ADE Desktop	Control plane for compact/continue; disk identity	Replace model weights
Compaction	Manage occupancy & drift	Be lossless for all artifacts (Factory: artifact trail stays weak)
Fertility tricks	Prefer dense formats / diffs	Guarantee higher task success

Context rot remains: larger windows under-attend mid-context. Compaction recovers **attention budget**, not omniscience.

Metacognition gap: models are weak at knowing *when* their own context is rotting unprompted — scaffolds must supply the rubric (SelfCompact).

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7. ADE mapping — today vs deep-dive bets

7.1 Shipped (spine)

Piece	Role
`ade.handoff/v1` + thrift Continuity	Tier-1 capsule + host `next_safe_command`
Soft system budgets (`context.rs`)	T0/rules/skills/handoff truncation
T0–T3 + `activate_skill`	Catalog > body (fertility + disclosure)
Effort = rounds + **output** tokens	Output discipline
Leases / goals / verify on disk	Tier-2 external memory seeds

7.2 Gaps vs state of the art

Gap	Research pointer
No automatic **tool-result clearing**	Anthropic context editing; Complexity Trap masking
No mid-thread **history** compaction @~70% / boundary	SelfCompact; Cursor; vendor compact APIs
Capsule sections not yet a hard checklist artifact trail	Factory artifact weakness
No fertility / compaction **gold** metrics	Measure mask vs summary vs capsule
No SelfCompact **rubric tool**	arXiv:2606.23525
chars÷4 unlabeled	Heuristic only

7.3 Recommended ADE roadmap (Token Usage / Compaction track)

Priority	Bet	Expected effect
C1	Tool-result clearing / observation masking policy	Biggest cheap win; often ≥ summary quality
C2	Structured capsule compaction at **task boundary** + safety @~70%	Continuity upgrade; less drift than threshold-only
C3	Write-before-compact: goals/decisions/verify → `.ade/` every phase	Protects artifacts Factory shows everyone loses
C4	SelfCompact-style `compact_context` tool + rubric	Adaptive timing without fine-tune
C5	Gold: dual-writer free; mask vs summary vs capsule fidelity	Stops cargo-cult ratios
C6	Fertility bench on ADE corpus (EN capsules, diffs, JSON)	Catches tokenizer surprises
Held	Private ADE cipher / stego language	Fails hard-limit test
Maybe	Open-weight compressor pair	Only if ADE hosts both ends

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8. Practical playbook (operators + builders)

Input

1. Catalog + IDs, not bodies.
2. Compact JSON / MD tables for structured state; pretty-print only offline.
3. Diffs/hunks, not full files.
4. English (or model's strongest coding language) for operator prose — don't chase exotic fertility.
5. Static → dynamic order (cache-stable).

Compaction

1. Clear tool blobs first.
2. Compact at **semantic boundaries**; threshold as backstop.
3. Structured sections: intent · decisions · paths · failing · next · verify.
4. Anchored merge > full regenerate when iterating capsules.
5. Externalize before summarize.

Output

1. Cap essays (Effort).
2. Prefer tool calls + short acts.
3. Verifiers emit pass/fail + pointers, not regenerated patches.

Never

1. Invent opaque ADE glyphs for “compression.”
2. Bill or optimize on chars÷4.
3. Mid-task model swap to “fix” occupancy.
4. Equate 99% compression ratio with success.

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9. One diagram

```
                    ┌─ Fertility (tokenizer) ── frozen on API ─┐
TEXT / DIFF/JSON ──┤                                            ├→ TOKENS IN WINDOW
                    └─ Compaction (harness) ── ADE can own ────┘
                                 │
             mask blobs · capsule · disk · Effort · Continuity
                                 │
                   invent private language? ── usually NO
```

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10. Conclusion

The real limits are clear:

- **LLMs** understand saturated languages and formats; they do not learn your cipher for free.
- **Tokenizers** set fertility; ADE cannot rewrite them on Claude/GPT.
- **Harnesses** own compaction — and that is where ADE should spend ambition.
- **Best “ADE language”** is already half-built: **`ade.handoff/v1` + skill IDs + diffs + path indexes`**, tightened into a write-before-compact loop with masking and boundary triggers.

Compaction research says: **mask early, structure the summary, externalize always, time the compact by trajectory — not by inventing a denser tongue.**

Appendix A — Selected primary sources

Compaction / agents

- Anthropic: *Effective context engineering for AI agents* (2025); Compaction docs; Context management (+editing/memory).
- OpenAI: Responses Compaction guide.
- Cursor: *Training Composer for longer horizons* / self-summarization (2026).
- Factory: *Evaluating Context Compression for AI Agents* (2025).
- arXiv:2508.21433 — *The Complexity Trap* (observation masking).
- arXiv:2606.23525 — *Self-Compacting Language Model Agents*.
- arXiv:2607.05378 — CompactionRL.
- arXiv:2510.00615 — ACON.
- LongLLMLingua — ACL 2024.
- MemAgent — arXiv:2507.02259.

Fertility / tokenization

- Petrov et al., NeurIPS 2023 — arXiv:2305.15425.
- Ahia et al., EMNLP 2023.
- Schmidt et al., EMNLP 2024 — *Tokenization Is More Than Compression* (arXiv:2402.18376).
- Zouhar et al., ACL 2023 — Rényi efficiency.
- Cognetta et al., LREC-COLING 2024 — Rényi gaming.

ADE internal

- `ADE-Tokenomics-IO-Context-Usage.md` · `TOKEN_ECONOMICS_RESEARCH.md` · `EFFORT_TURN_BUDGET_PLAN.md` · `crates/core/src/handoff.rs` · `crates/agents/src/context.rs`

Appendix B — Confidence caveats

- Factory vendor ranking is first-party — treat as directional.
- Code fertility table is **MEDIUM**; re-measure on ADE corpus.
- LongLLMLingua/MemAgent transfer to full tool agents is **MEDIUM**.
- SelfCompact numbers are paper-reported; dogfood on ADE gold before product claims.

End · `ade.compaction-research/v1` · *2026-07-23*